A Cross-Subject sEMG-to-Speech Conversion System Using Content Features and Model Calibration

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***Abstract*— Current sEMG-based speech generation methods primarily rely on large-scale datasets from single participants, which imposes a burden on users. Moreover, previous research methods often require synchronous recordings of sEMG speech for model training, making them unsuitable for patients with speech impairments. This article introduces a cross subject sEMG-to-speech (ETS) conversion system based on content features and model calibration methods. This method uses a pre trained acoustic model to extract speaker independent acoustic features, and adapts the model to the EMG features of new individuals through Child Tune model calibration. In order to develop an ETS system suitable for people with language barriers, we recommend using electronic synthesized audio to train the ETS model as a substitute for human voice, and reconstructing speech through a voice encoder with speech conversion function. The experimental results show that our proposed sEMG-to-speech (ETS) conversion system can achieve a CER of 21.71% after calibration with 20 minutes of data from new users, and using electronic synthesized audio to train the ETS model can achieve a CER equivalent to using human voice.**

***Index Terms*— Silent speech interface, speech synthesis, model calibration, electromyography.**

I. Introduction

LANGUAGE constitutes a fundamental mode of communication among humans; nevertheless, not all individuals possess typical language communication capabilities. For example, people who have undergone laryngectomy surgery or individuals with articulation disorders may lose their ability to speak. Several studies have shown that the prevalence of this disability is significant [1], [2]. In the absence of clinical procedures to repair damage, silent speech interface (SSI) is considered a promising device which can be used to restore communication abilities for individuals with speech impairments.

SSI is an assistive device that decodes the non-vocal biological signals produced during the generation of speech to restore oral communication [2], [3]. Currently, various sensing modes are used to study capturing speech-related biosignals and decoding speech content from them [4], [5], [6], [7], [8], [9], [10]. Among them, surface electromyography(sEMG) can non-invasively record facial and neck muscle bioelectric signals during speech production. This property makes sEMG-based SSI an ideal method for individuals with intact muscle function in their speech pronunciation apparatus, such as healthy subjects or patients who have undergone laryngectomy surgery.

Early research on silent speech interfaces based on sEMG signals dates back to the mid-1980s, with most studies focusing on extremely limited vocabularies comprising only a few words or commands [6], [11], [12], [13], [14]. With the development of deep learning models, many studies have recently attempted to directly reconstruct speech from sEMG. In 2016, Diener et al. gathered audible audio data and corresponding sEMG recordings, achieving for the first time, speech reconstruction from sEMG using deep neural networks (DNN) [15]. Subsequently, Gaddy and Klein introduced the Transformer model, aligning the sEMG signals of spoken and silent speech with the Dynamic Time Warping (DTW) algorithm, attaining speech synthesis from silent speech sEMG signals with a word error rate of 42% [16]. In 2023, Scheck et al. developed a low-latency sEMG to Speech system [17]. Recently, Li et al. first achieved the reconstruction of Mandarin in sEMG signals [18]. Despite notable advancements, sEMG-to-Speech synthesis technology continues to encounter challenges:

Firstly, it can be observed that most of the currently reported sEMG-Speech methods use large datasets from single participants. For example, Gaddy et al. collected a dataset lasting 18.6 hours from one subject. However, the prolonged collection of sEMG signals imposes a significant burden on participants. Additionally, for multiple trials collected from one participant, due to variations in electrode placement and

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Ethics Committee of Guangzhou First People's Hospital, School of Medicine, South China University of Technology under Application No. K-2019-064-01.

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fluctuations in skin impedance, models trained on previous trial data still exhibit performance degradation when faced with new trial data. Current literature mainly focuses on single-participant sEMG-Speech methods. Due to the large inter-subject variability of sEMG signals, i.e., models may be overly tailored to the specific participant they were trained on, their generalizability to new participants might be limited. Unlike classification from sEMG signals to predict text, cross-participant sEMG-Speech methods are more challenging due to audio specificity caused by differences in timbre among different participants. To address this, Scheck et al. used a pre-trained Hubert model to extract audio content features from audio as target features, thereby reducing the specificity of the prediction targets.

Additionally, current sEMG-to-Speech technology requires the sEMG signals and synchronous vocal audio signals from users while speaking, to train the transduction model that convert sEMG signals feature into audio feature [7], [8], [16], [17], [18], [19]. However, this approach is unsuitable for individuals with language impairments who have lost their ability to speak, such as those who have undergone laryngectomy surgery to treat laryngeal cancer.

To address this issue, Li et al. [20] proposed a data collection scheme in which individuals with speech impairments act as silent speakers and healthy individuals as auxiliary speakers, simultaneously reading the same text. The sEMG signals from the silent speakers are converted into the auxiliary speakers' audio. While this method shows promising results, it may cause inconvenience and only allows reproduction of the auxiliary speakers' voices.

In this study, we propose a novel cross-subject sEMG-to-speech (ETS) conversion system. Specifically, a new framework, termed WE-ETS, is developed to establish a robust mapping between sEMG signals and corresponding speech outputs. The proposed system leverages the pre-trained Whisper model to extract Whisper-based audio embeddings, which serve as high-level speech features. These embeddings have demonstrated robustness and effectiveness in prior studies on voice conversion [21] and speech decoding from electrocorticography (ECoG) [22], [23], supporting their applicability to sEMG-based speech reconstruction.

To enhance the adaptability of the proposed system to new subjects, we employed the Child-Tuning method proposed in [24] for model calibration.

Furthermore, we designed and implemented a dedicated data collection protocol specifically for individuals with speech impairments. This protocol facilitated a systematic evaluation of the effectiveness of ETS systems integrated with speech conversion capabilities in restoring vocal communication functions.

II. Data Acquisition

A. Recording Hardware

The sEMG signal acquisition device developed in this study is shown in Fig. 1. The device is equipped with two 16-channel electronic skins to collect sEMG signals of the facial and mandibular muscle groups, respectively. The electronic skin

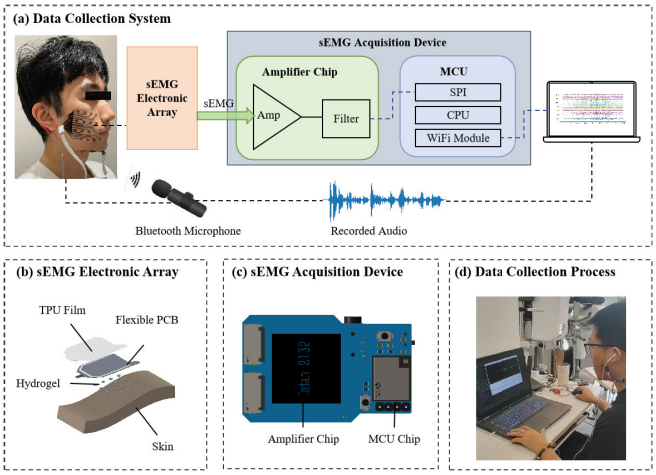

Fig. 1. The sEMG signals and audio signals collection process. The sEMG signals are collected by a wearable 32-channel sEMG signal acquisition device and two electronic skins attached to the face and jaw. These electronic skins are constructed from flexible PCB, hydrogel, and medical tape. The collected sEMG signals are transmitted to the computer through a WiFi module. Audio signals are recorded and transmitted to a computer through a Bluetooth microphone.

is composed of a hydrogel layer, a flexible PCB layer, and a TPU film layer. A dense array of metal electrodes ensures high spatial resolution of the sEMG recordings. Each electrode is coated with conductive hydrogel to reduce skin-electrode impedance and mitigate deformation-induced artifacts. The collected sEMG signals are transmitted from the electronic skin through a ribbon cable to an amplifier module based on the RHD2132 chip for amplification and analog-to-digital conversion. Then, the ESP32 chip is used as the MCU and the WiFi module. The acquisition circuit applies a 0.5 Hz-500 Hz band-pass filter, then transmits the digitized sEMG data via WiFi to a computer for storage. Simultaneously, a Bluetooth microphone records audio in sync with the vocal sEMG.

B. Dataset Information

We recruited a cohort of 11 healthy volunteers, all native Mandarin speakers with standard oral expression abilities, including 9 males and 2 females, with a mean age of 25 years. Informed consent was obtained from all participants prior to their involvement in the study. Prior to sensor placement, participants cleaned the facial and chin skin with a scrub agent and alcohol swabs to remove sebum. The experimenter then attached the electronic-skin patches to the facial and chin areas. Participants sat before a computer running the sEMG acquisition software and read on-screen sentences in vocalized and silent modes. Sentences were repeated three times silently and one to three times vocally.

The dataset corpus was designed to ensure phoneme balance and multi-scenario coverage, comprising approximately 300 sentences drawn from daily conversations, human-computer interaction commands, and written texts such as news and novels. Additionally, the corpus includes declarative, interrogative, and imperative structures to enhance grammatical diversity.

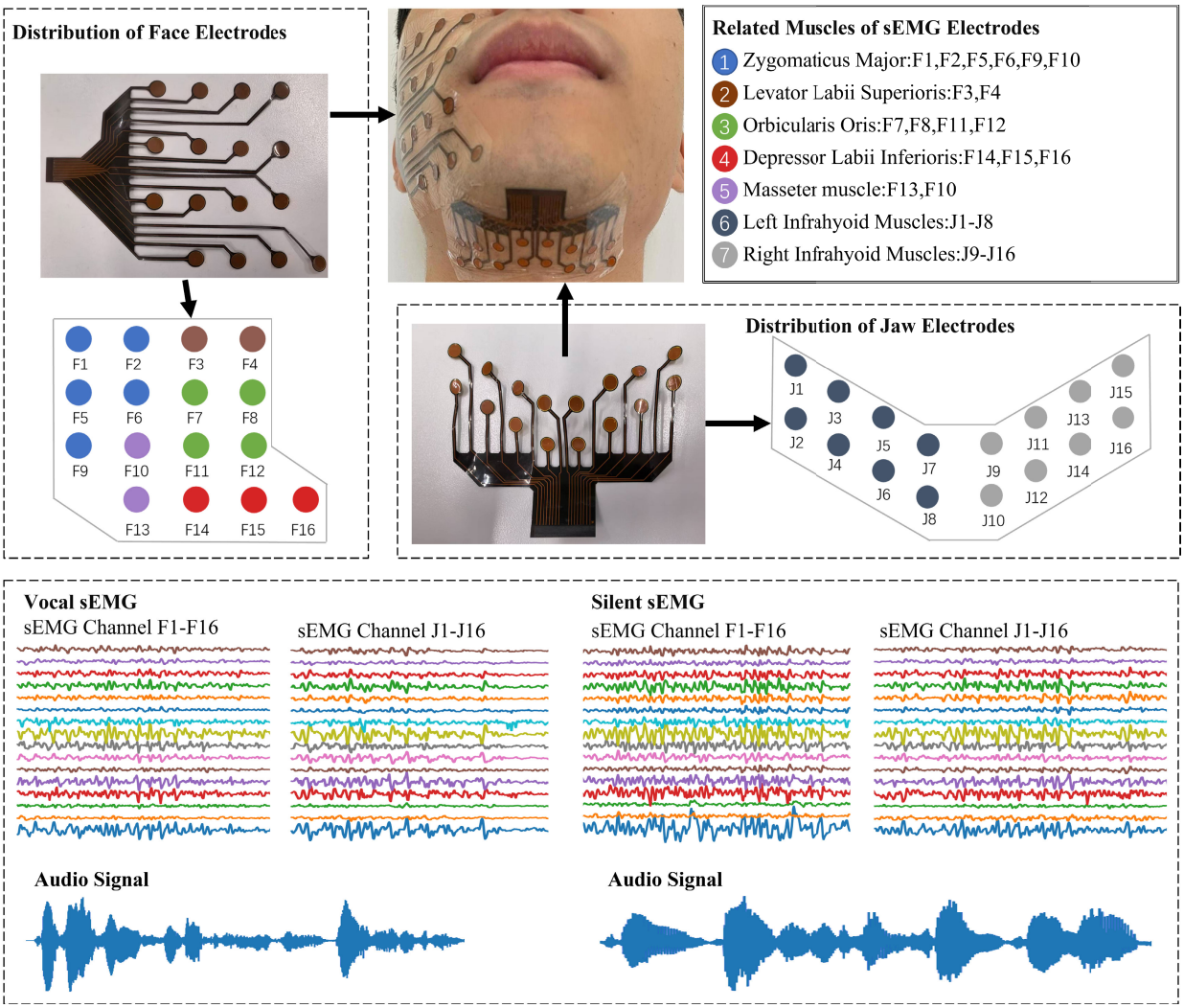

Fig. 2. The distribution and related muscles of the sEMG electrode on the jaw and face. We use electrodes of different colors to indicate the muscle group from which the electromyographic signal is collected. We select sEMG signals collected in silent and vocal modes and their corresponding audio signals, and plot their waveform.

During the silent mode, while the sentence was displayed, the computer concurrently played back the electronic synthesized speech of that sentence at a consistent and unhurried pace. The subjects were directed to mimic the electronic voice during silent reading, permitting facial and cervical muscle movements without vocalization. The surface sEMG signals generated during this process were captured by the sEMG collection device.

In the vocal mode, the participants were obliged to vocalize the sentences shown on the computer monitor aloud, with the sEMG collection apparatus and a Bluetooth microphone concurrently recording the synchronized sEMG and audio signals.

Our sEMG-to-Speech dataset information for each subject is shown in Table I. We ensured a minimum data collection duration of 40 minutes per subject, with the entire data collection experiment spanning approximately 3-4 hours for each participant. Figure 2 illustrates the sEMG signals and corresponding audio captured during the reading of a sentence in both modes.

It is noteworthy that previous studies [16], [17], [18], [20], [25], [26] typically utilized both vocal and silent sEMG recordings for model training. In contrast, this chapter adopts a different silent sEMG collection paradigm, wherein participants were instructed to silently imitate computer-played audio at a consistent speech rate. Furthermore, the feasibility of training models using only the playback audio and silent sEMG signals is further explored in subsequent chapters. Although this represents a minor modification, it offers practical advantages for advancing the application of SSI systems in individuals with speech impairments.

This study was reviewed and approved by the Ethics Committee of Guangzhou First People's Hospital, School of Medicine, South China University of Technology (Protocol No. K-2019-064-01).

C. Data Processing

Our sEMG signal acquisition system recorded the raw sEMG signals from 32 channels at a sampling rate of 1 kHz. After completing the collection of the original sEMG data,

TABLE I
STATISTICS OF OUR sEMG-TO-SPEECH DATASET

speaker ID	Silent Speech Time(Min)			Vocal Speech Time(Min)	Number of Utterances		Total time(Min)	Total Utterances
	Train	Val	Test		Silent	Vocal		
S1	35.5	3.56	3.31	43.67	408	423	86.04	831
S2	26.45	3.58	3.73	27.53	303	300	61.29	603
S3	20.32	3.14	3.27	22.97	226	224	49.70	450
S4	40.13	3.34	3.26	12.47	289	101	59.20	390
S5	81.8	3.41	3.68	28.98	595	256	117.87	801
S6	34.08	3.46	3.73	23.67	246	245	64.94	491
S7	47.06	3.07	3.53	28.32	450	267	81.98	717
S8	48.32	3.33	3.17	27.47	457	306	82.29	763
S9	48.17	3.45	3.23	28.41	457	305	83.26	762
S10	47.2	3.37	3.01	27.78	456	305	81.36	761
S11	29.51	3.12	3.88	18.43	347	231	54.94	578

further data preprocessing is required. We employed the Dobesch wavelet transform of order 8 for the enhancement of physiological electrical signal feature extraction. Additionally, a notch filter was utilized to eradicate the interference from the 50 Hz power frequency and its associated harmonics.

The dataset comprises sEMG signals and speech signals captured during both vocal and silent speech conditions. This includes acoustic sEMG and synchronously recorded vocal speech, as well as silent sEMG and corresponding synthesized sounds. While acoustic sEMG and vocal speech are deemed synchronous recordings, sound signals typically lag behind sEMG signals due to electromechanical delay effects. To address this, a 50ms delay compensation is employed to ensure temporal alignment between voice and sEMG. Additionally, the text of each utterance is preserved and will serve as a reference for evaluating the intelligibility of the synthesized speech. The data collected during the silent mode from each subject, amounting to approximately 2 minutes, will be partitioned into test and validation sets.

To obtain the frame-based phoneme labels for the target audio, we use the Montreal Forced Aligner [27]. The aligner uses an ASR model trained on a Mandarin dataset to recognize the phoneme sequence of the audio at the frame level and obtains the frame-based phoneme label according to the phoneme dictionary.

III. METHODS

A. Content Feature

Whisper is a robust automatic speech recognition (ASR) model that employs an encoder-decoder architecture. Numerous studies in various domains have validated the robustness and expressiveness of the content features produced by Whisper encoders. In this paper, we adopt a pre-trained Whisper-medium model encoder as the acoustic content encoder [28], as depicted in Fig 3, to generate target content features, named Whisper-Embedding. We selected Whisper model because its multilingual training (which includes Mandarin) allows it to capture fine-grained linguistic information, including tonal patterns, from speech signals.

B. Pipeline of Proposed ETS System

Our WE-ETS system comprises three components: a feature transduction model, a HiFiGAN vocoder, and a speaker encoder. The feature transduction model is trained using sEMG dataset we collected and the audio content feature extracted by pre-trained SSL models and Whisper model encoder.

The training and inference pipeline of our ETS system are illustrated in Fig 3. The primary objective during the training process is to develop a transduction model capable of converting sEMG signals into speech content features.

For purposes of comparison and illustration, the architecture of the feature transduction model employed herein is identical to baseline, which comprises three one-dimensional (1D) convolutional residual blocks and six Transformer layers with relative position embedding.

During the training process, the feature transduction model accepts sEMG signals as input and produces predicted phonemes along with content features. The model is trained by minimizing the discrepancy between the predicted and target content features, utilizing an auxiliary frame-level phoneme loss to facilitate the training process. Subsequently, the target speech is fed into the acoustic content encoder to extract the target content features.

To efficiently process sEMG sequences of varying lengths during training, we employ a sampling strategy that groups a batch of sequences with a cumulative length of approximately N. These sequences are then concatenated and reshaped into a batch-fixed-length length S for input into the network. Consequently, the effective batch size for the training model becomes N/S. For the audio content features predicted by the model, alignment and loss calculations are based on recording sequences of opposing original lengths.

During the inference process, the trained feature transduction model is employed to forecast the content features of the target speech from the sEMG signal features. Concurrently, the mel spectrum of the reference speech is fed into the pre-trained speaker encoder to acquire speaker embeddings. Finally, these speaker embeddings, along with the predicted content features, are inputted into the pre-trained HiFiGAN-based vocoder. The resultant synthesized speech generated by this model mirrors the content of the target speech, while the speaker's identity is emulated from the reference audio.

C. Speaker Encoder and Vocoder

The pre-trained HiFiGAN vocoder and speaker encoder for content features extracted by Whisper and wav2vec are derived

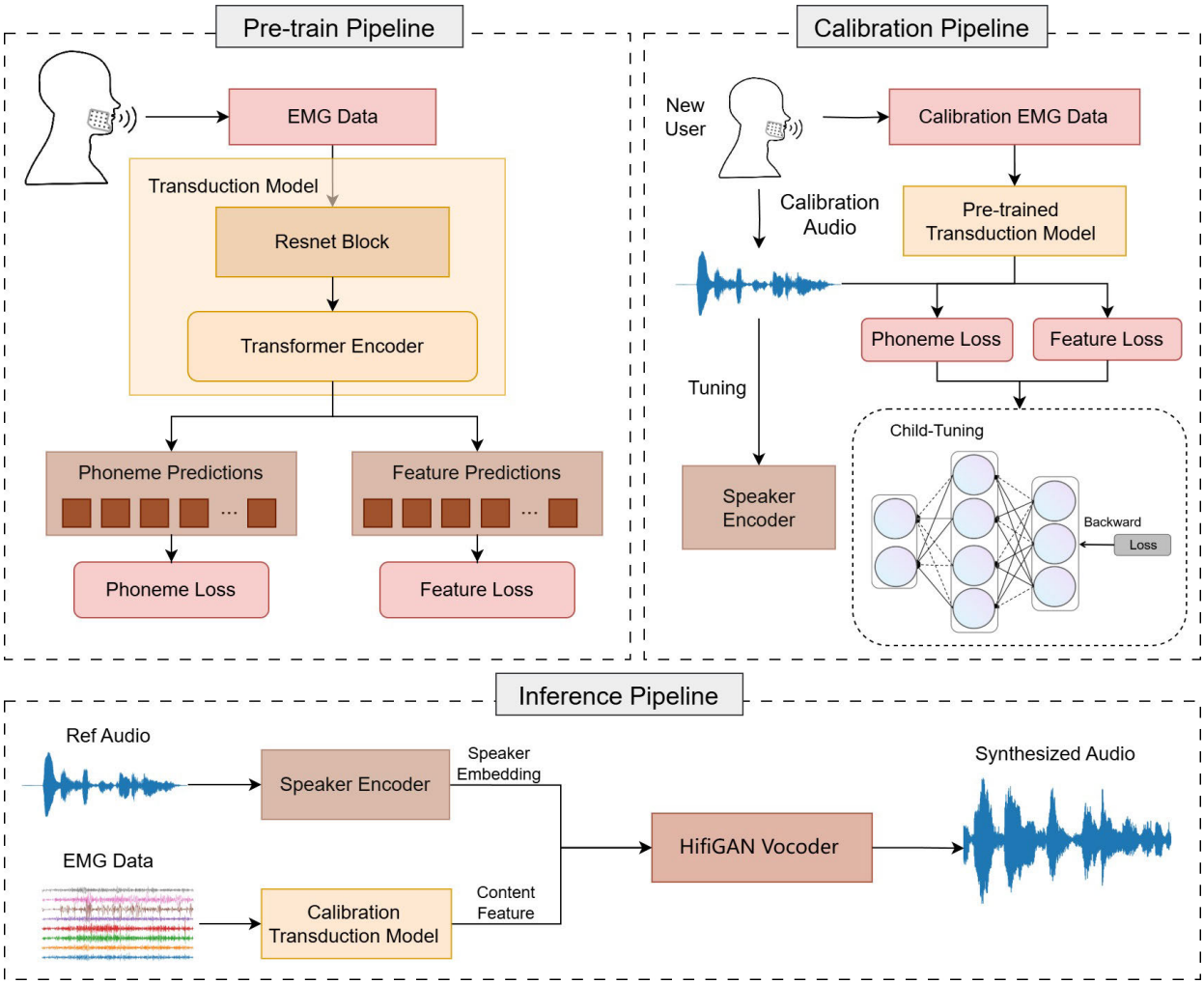

Fig. 3. The pre-train, calibration, and inference pipeline of our proposed WE-ETS system.

from ConsistencyVC [29]. This VC system, inspired by the architecture of FreeVC-S [30], employs speaker consistency loss to collaboratively train the speaker encoder. The HiFiGAN vocoder and speaker encoder for HuBERT are trained using the Aishell3 dataset [31] and the code provided by ConsistencyVC.[1]

During inference stage, the speaker encoder transforms the variable-length mel-spectrogram of a reference speech into a fixed-size speaker embedding. Subsequently, the HiFiGAN vocoder utilizes both the speaker embedding and predicted content features for the reconstruction of speech.

D. Alignment and Loss Function

The feature transduction model is trained using the sEMG dataset we collected. Our sEMG dataset consist of the sEMG data collected in vocal mode and silent mode, and the time-align audio collected in vocal mode. The sEMG signals and target audio are divided and feature extracted with the same frame shift of 20ms and frame length of 80ms.

For the sEMG data collected in the vocal mode, its target audio is the synchronously collected audio A_v. At this time, sEMG signal sequence E_v has the same length as the audio content feature sequence C_v, and each frame of vocal sEMG input $E_{v[t]}$ corresponds to the same frame of time-aligned audio content feature $C_{v[t]}$, as well as the phoneme label $P_{v[t]}$. For the synchronously collected sEMG and audio data in the vocal mode, the content feature loss $\mathcal{L}_{cv}$ is the mean distance between target and aligned predicted content feature:

$$\mathcal{L}_{cv} = \frac{1}{T}\sum_{t=1}^{T} \|C_{v[t]} - \hat{C}_{v[t]}\|_2 \tag{1}$$

where $C_{v[t]}$ is the target content feature and $\hat{C}_{v[t]}$ is the predicted content feature at frame t. The phoneme classification loss $\mathcal{L}_{pv}$ is the cross-entropy loss between the target phoneme and the aligned predicted phoneme:

$$\mathcal{L}_{pv}(E) = -\frac{1}{T}\sum_{t=1}^{T}\sum_{i=1}^{|P_v|} b_{t,i} log\, \hat{P}_v(\mathbf{x})_{t,i} \tag{2}$$

where Pv is the set of phoneme classes, $\hat{P}v$ is the sequence of phoneme predictions of length T, and $b_{t,i}$ is the one-hot vector of target phoneme class i at frame t.

The total loss of the feature transduction model is a weighted sum of the content feature loss $\mathcal{L}_{cv}$ the phoneme

[1]https://github.com/ConsistencyVC/ConsistencyVC-voive-conversion

classification loss $\mathcal{L}_{pv}$:

$$\mathcal{L}_{total} = \lambda_c \mathcal{L}_c + \lambda_p \mathcal{L}_p \tag{3}$$

where λ_c and λ_p are the weights of $\mathcal{L}_{cv}$ and $\mathcal{L}_{pv}$ respectively.

For the sEMG data collected in silent mode, there is an absence of synchronous audio as the target speech. During the data collection process, each sentence of text was read and recorded in both vocal and silent modes. Consequently, for sEMG recorded in silent mode, we could utilize audio recorded in vocal mode with the same text as the target speech. Since the silent sEMG E_s and audio A_v were recorded separately, the loss calculation between the target results from A_v and predicted results from E_s require an alignment between the two recordings. As Gaddy and Klein [25], the alignments in each training step are formed by DTW. DTW is a dynamic programming algorithm utilized to identify the optimal monotonic alignment between two sequences at minimal cost. The cost matrix δ is composed of the paired euclidean distances between target content feature $C_{v[t]}$ from A_v and the predicted content feature $\hat{C}_{s[t]}$ from E_s, as well as the phoneme negative log likelihood:

$$\delta[i, j] = \lambda_c \left\| C_{v[i]} - \hat{C}_{s[j]} \right\|_2 - \lambda_p P_{v[i]}^{\top} \log \hat{P}_{s[j]} \tag{4}$$

where $\delta[i, j]$ is the cost between target frame i and source frame j, $\hat{C}_{s[j]}$ and $\hat{P}_{s[j]}$ are the predicted content feature and phoneme of frame j, and $C_{v[i]}$ and $P_{v[i]}$ are the source content feature and phoneme of frame i from A_v

In addition, considering that individuals with speech impairments may only be able to collect data in silent mode, we also contemplated using electronic audio A_e followed by participants in silent mode as the target speech. When using A_e as the target speech, the cost matrix is:

$$\delta[i, j] = \lambda_c \left\| C_{e[i]} - \hat{C}_{s[j]} \right\|_2 - \lambda_p P_{e[i]}^{\top} \log \hat{P}_{s[j]} \tag{5}$$

where $C_{e[i]}$ and $P_{e[i]}$ are the source content feature and phoneme of frame i from A_e.

DTW employs dynamic programming to identify the optimal warping path, which minimizes the cost matrix δ. The alignment matrix a, derived from this optimal warping path, maps the predicted content feature $C_{[i]}$ of each frame to the target content feature $\hat{C}_{[}j]$ of a single frame. The total loss of the content feature predicted from E_s is the mean of aligned pairwise distances:

$$\mathcal{L}_{total} = \frac{1}{N} \sum_{i=1}^{N} \delta[i, a[i]] \tag{6}$$

E. Model Calibration

To better adapt the pre-trained sEMG-Speech model to new users and sessions, we fine-tune the model using the Child-Tuning method. This method suggests that there is no need to update all parameters during calibration, especially for large models. This fine-tuning approach obtains the parameters that need to be updated through both task-driven and non-task-driven methods and applies a mask to the parameters that are not updated during backpropagation [24]. Specifically, we introduce a gradient mask during the backward process to update a subset of parameters (i.e., sub-network), while still utilizing the entire model's parameters in the forward process.The vanilla process of backpropagation for fine-tuning can be described as follows:

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \frac{\partial \mathcal{L}(\mathbf{w}_t)}{\partial \mathbf{w}_t} \tag{7}$$

where $\partial \mathcal{L}(\mathbf{w}_t)$ are the gradients corresponding to the model parameters $\mathbf{w}_t$, and η is the learning rate.

Child-Tuning decides a sub-network at each iteration and updates only those parameters through masking. The back-propagation formula is as follows:

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \frac{\partial \mathcal{L}(\mathbf{w}_t)}{\partial \mathbf{w}_t} \odot M_t \tag{8}$$

where M is a 0-1 mask matrix, which is defined by the following formula:

$$M_t^{(i)} = \begin{cases} 1, & \mathbf{w}_t^{(i)} \in \mathcal{C}_t \\ 0, & \mathbf{w}_t^{(i)} \notin \mathcal{C}_t \end{cases} \tag{9}$$

1) Task-Free Child Network Detection: Task-Free Child Network Detection is a method for selecting subnetworks without requiring any downstream task data. Specifically, in each iteration, a binary mask M is generated based on a Bernoulli distribution with probabilityp_F:

$$M_t \sim \text{Bernoulli}(p_F) \quad (4) \tag{10}$$

When p_F equals 1, it degenerates into the ordinary tuning method.

Child-Tuning can be understood as a potent regularization strategy for the optimization process. It facilitates the model's ability to circumvent saddle points within the loss landscape and promotes convergence towards a flatter local minimum, thereby enhancing stability and generalization capabilities.

2) Task-Driven Child Network Detection: Task-Driven Child Network Detection identifies the most critical subnetworks for a specific downstream task. Specifically, it employ Fisher Information Estimation to identify subsets of parameters that are highly correlated with the given task. The Fisher Information Matrix (FIM) pertaining to the model parameters w is formally defined as follows:

$$\mathbf{F}(\mathbf{w}) = \mathbb{E}\left[\left(\frac{\partial \log p(y|\mathbf{x}; \mathbf{w})}{\partial \mathbf{w}}\right)\left(\frac{\partial \log p(y|\mathbf{x}; \mathbf{w})}{\partial \mathbf{w}}\right)^{\top} \right] \tag{11}$$

. Where x and y denote the input and output, respectively. This can also be viewed as the covariance of the gradient of the log likelihood with respect to the parameters w.

To reduce computational overhead, we derive the sub-adjustment child network only at the beginning of fine-tuning and keep it unchanged throughout the fine-tuning process.

IV. Validation Experiment and Result

To assess the efficacy of content features extracted by the Whisper model, we employed the Direct ETS model from [25] as a benchmark, utilizing MFCCs as its target features. To ensure a fair comparison, our implementation featured a Transformer encoder identical in architecture and dimensions to the baseline model. Given the discrepancy in

frame rates between MFCCs and Whisper features, the ResNet configuration in Direct ETS comprised three layers of one-dimensional convolutions with a stride of 2 followed by a single layer with unit stride. In contrast, the WE-ETS ResNet architecture consisted of four such convolutional layers, each with a stride of 2.

When evaluating cross-subject calibration methods, we employed leave-one-out cross-validation to train a pre-trained model. Subsequently, we fine-tuned and tested the model using the data from the target subject. We investigated the impact of Child-Tuning, recalculating training from scratch, and traditional fine-tuning (normal tuning) on calibration datasets of varying sizes. Additionally, to assess the effect of training the WE-ETS model with electronic audio as the target audio, we conducted experiments using four different data settings.

A. Model Parameters Settings

In the baseline Direct ETS system, sEMG signals are extracted with a frame length of 27 ms and a frame shift of 10 ms. In our ETS system, Whisper employ a hop length of 320 and a window length of 1024 with a sampling rate of 16000 Hz for content feature extraction. To accommodate these extracted acoustic content features, we modified the vocoder and feature transduction model. The amplification sequence for the four residual blocks in the HifiGAN-based vocoder is specified as [10,8,2,2], and kernel sizes of [20, 16, 4, 4] are used to avoid potential checker-board artifacts [32]. To align the content feature sequence, we extracted sEMG signal features at 800Hz with a hop length of 16 and a window length of 64. Additionally, we ascertain the output dimension of the feature transduction model and the input dimension of the vocoder model, guided by the dimensions of the content features.

To ensure the validity of the experiment, sEMG data with a cumulative duration exceeding three minutes were randomly chosen from the silent mode recordings obtained for each participant. These selections served as the testing and validation set, ensuring that they had no overlap with the training set.

In the training stage of feature transduction model, we set the total sequence length $N = 16000$ (320 seconds), the sequence length $l = 50$ (1 second), the epoch $= 180$. The learning rate is 0.002 with a linear warm up of 500 iterations. The determined weight between $\mathcal{L}_c$ and $\mathcal{L}_p$ are set to 1 and 0.5, respectively. All hyperparameters are determined using the grid search method.

Followed by [25], the learning rate schedule with weight decay $= 10^{-7}$, reduce the learning rate by half when the validation loss do not improve for five consecutive epochs. The Adam optimizer is used for model training.

B. Evaluation Metrics

To objectively assess the quality of synthesized speech, we employ Mel-Cepstral Distortion (MCD) [33] to gauge its distortion and comprehensibility. MCD is calculated as follow:

$$\mathrm{MCD[dB]} = 10/\ln 10 \sqrt{2 \sum_{n=1}^{N} \left(\hat{d}_n - d_n\right)^2} \tag{12}$$

where $\hat{d}_n$ and d_n represent the n-th dimension of Mel Cepstral Coefficients derived from synthesized and ground truth audio samples, respectively. A lower MCD value indicates minimal distortion in the converted sound.

Following [34], we adopt F0 Pearson correlation coefficient (F0CORR) as an evaluation metric to assess the tonal variation of the generated speech.The value of F0CORR ranges from 0 to 1, with higher values indicating a greater similarity in tonal variation between the synthesized speech and the target speech.

Besides, We utilize Automatic Speech Recognition (ASR) models to transcribe both synthesized and actual speech, subsequently calculating Character Error Rate(CER) as an objective indicator for speech quality evaluation. The ASR model employed is a conformer model based on PaddlePaddle framework, trained on approximately 10000 hours of the WenetSpeech dataset. The predicted text is derived through a beam search decoding process.

To conduct a subjective evaluation of the synthesized speech quality, we recruited 18 native Mandarin-speaking individuals with normal hearing to conduct listening tests. From synthesized audios from each model, 20 sentences are Randomly selected and presented in a randomized order. The participants were tasked with rating the naturalness of each sentence and its resemblance to the reference audio on a Likert scale, ranging from 1.0 to 5.0. We calculated the mean opinion scores for naturalness(NMOS) from all participants to serve as the subjective evaluation metric for the synthesized speech.

Simultaneously, to gauge the speech conversion functionality of our WE-ETS system, to evaluate the speech conversion functionality of our WE-ETS system, we employed the Mean Opinion Score for the similarity (SMOS) from the listening test as a subjective evaluation metric. Additionally, a Speaker Verification (SV) system was utilized to classify the synthesized speech using both seen and unseen audio samples as references, in order to verify if the speaker matched the reference audio.

C. Comparison of ETS System

We utilized the Direct sEMG to Speech (Direct ETS) system, as proposed by Gaddy and Klein [25], as a baseline for comparison. This system trains a feature transduction model to directly predict MFCCs of the target speech from sEMG signals and employs a HiFi-GAN vocoder to synthesize speech from these MFCCs.

To ensure comparability, we adopted the same architecture for our feature transduction model as that used in the Direct ETS system. We initially evaluated both our proposed WE-ETS system and the baseline ETS system when training the model with on full data set from each subject.

Consistent with prior research [17], [18], [25], [26], we utilized the E_v and E_s, collected in vocal and silent modes respectively, to jointly train the feature transduction model. Here, E_v was trained using the time-aligned audio A_v as the direct training target, whereas E_s was trained through alignment with A_v via a dynamic warping algorithm. The vocoder of our WE-ETS system employed the predicted

TABLE II
EVALUATION OF THE SYNTHETIC SPEECH FROM BASELINE SYSTEM AND OUR PROPOSED WE-ETS SYSTEM

System	Metrics	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	Mean	Std
Direct ETS	MCD(dB)	6.98	6.21	6.74	6.01	5.72	5.97	6.22	6.28	6.37	6.65	6.93	6.37	0.389
	CER(%)	26.48	31.43	12.76	38.37	47.97	33.73	42.14	25.19	18.92	16.21	36.81	30.45	10.906
	F0CORR	0.358	0.324	0.417	0.301	0.279	0.241	0.315	0.287	0.436	0.379	0.284	0.329	0.0584
	NMOS	2.13	2.27	1.35	1.15	2.46	1.98	1.24	2.34	2.69	2.81	1.23	1.97	0.592
WE-ETS	MCD(dB)	4.76	4.89	4.65	4.72	5.48	4.96	4.38	4.65	4.51	4.94	4.58	4.77	0.281
	CER(%)	5.41	6.85	7.64	2.58	5.76	3.58	3.46	2.81	1.55	2.63	11.19	4.86	0.273
	F0CORR	0.593	0.493	0.482	0.588	0.516	0.531	0.614	0.652	0.693	0.674	0.468	0.573	0.0762
	NMOS	3.89	4.31	3.58	3.88	3.55	3.72	3.37	3.36	4.34	3.47	3.46	3.72	0.332

Whisper-Embedding features alongside the subject's own speech as a reference for speech synthesis.

Tables II encapsulate the evaluation of synthetic speech quality for the baseline ETS system and the WE-ETS system across all subjects, presenting the average values obtained.

As indicated in Table II, synthetic speech from WE-ETS outperforms the baseline in subjective NMOS and objective metrics (MCD, CER, F0CORR). The CER of synthesized speech using our proposed WE-ETS system from predicted Whisper-Embedding features is 4.86%, which is significantly lower than the CER of synthesized speech using the Direct ETS system (30.45%). This confirms the superior performance of our WE-ETS system on the low-resource dataset for the sEMG-to-Speech task. The WE-ETS system achieved an average F0CORR of 0.573, which was significantly higher than the F0CORR of 0.329 obtained by the Direct ETS system, demonstrating the superior capability of WE-ETS in predicting tonal variations.

A plausible explanation for the differences in the quality of the generated speech is that the features extracted from the pre-trained Whisper model prioritize the capture of semantic information contained in the utterances, thereby exhibiting greater robustness and tolerance to ambiguity and noise, and reducing the need for overfitting in the feature transduction model. Furthermore, the use of weakly supervised training on large-scale datasets encompassing multiple languages enables Whisper to incorporate extensive prosodic and stress information. This may enhance its suitability for modelling tonal languages such as Mandarin.

Among the systems evaluated, the Direct ETS system achieved the CER of 30.45% on synthetic speech in our dataset. This performance is comparable to the results reported by Li et al. [20] for the same system on synthetic speech in Mandarin low-resource datasets. Notably, when compared to the research [20] that employed mel-spectrograms as target features, our WE-ETS system have achieved better outcomes in CER but recorded a higher MCD. This discrepancy may be due to our use of Whisper-Embedding as the target audio feature, which prioritizes the semantic content of speech over its spectral characteristics. In the context of the sEMG-to-Speech task, synthesizing speech with accurate content from sEMG data is our primary concern.

D. Cross-Subject Calibration Evaluation

We utilize training data from various speakers to develop a cross-speaker WE-ETS model. The model is fine-tuned using varying scales of absent speaker data, referred to as model calibration. For comparison, we also train models from scratch with the same scale of data used for calibration. Additionally, we compare direct data fine-tuning (Normal Tuning) with Task-Driven (Child Tuning-D) and Task-Free (Child Tuning-F) Child-Tuning methods.

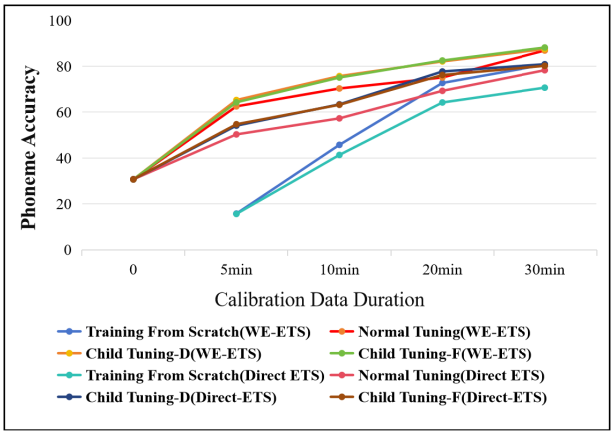

Fig. 4. The Cross-subject Phoneme Accuracy of WE-ETS System under Different Calibration Conditions.

Fig 4 and Fig 5 present the cross-subject average CER and phoneme classification accuracy under different calibration conditions. The results indicate that the fine-tuning methods significantly outperform the model trained directly from scratch, suggesting that the pre-trained model's knowledge can be partially transferred to new users. Furthermore, the Child Tuning method demonstrates superior performance compared to Normal Tuning, particularly when the fine-tuning data is limited, where this advantage is more pronounced.

A comparative analysis between the Direct ETS and WE-ETS models reveals that the latter demonstrates superior performance in both Character Error Rate (CER) and phoneme accuracy when applied to new subjects, utilizing identical calibration methodologies and equivalent volumes of calibration data. This enhanced performance substantiates the hypothesis that incorporating speaker-independent content features effectively mitigates the individual-specific characteristics inherent in model training objectives. Consequently, this approach significantly bolsters the model's generalization capabilities across diverse subject populations.

E. WE-ETS System Trained Without Vocal Audio

We assess the efficacy of WE-ETS models under four distinct data configurations:

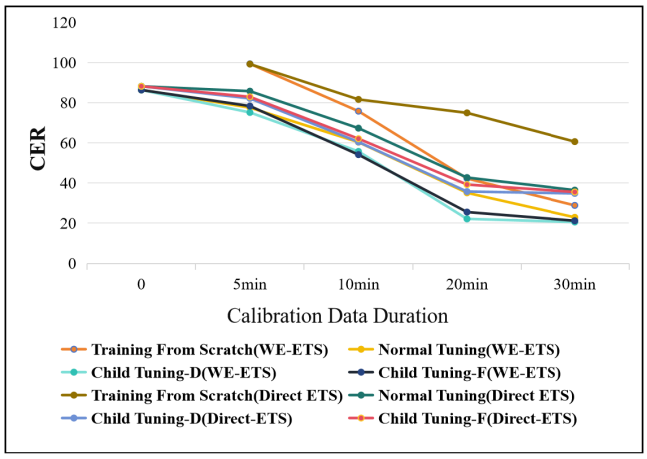

Fig. 5. The Cross-subject CER of synthesized speech synthesized by WE-ETS System under Different Calibration Conditions.

TABLE III
EVALUATION OF THE SYNTHESIZED SPEECH IN THE WE-ETS SYSTEM UNDER VARYING TRAINING DATA CONDITIONS

Data Configurations	CER	NMOS	SMOS	SV accuracies	
				Seen	Unseen
Full-A_v	4.86	3.72	3.62	98.6	80.7
Full-A_e	6.54	3.48	3.21	96.4	78.6
Silent-A_v	6.35	3.63	3.65	98.2	80.2
Silent-A_e	8.46	3.31	3.25	95.5	77.3
GT Audio	0.77	4.78	4.11	94.3	79.8

1) Full-A_v: utilizing the recorded voice A_v in audio mode as the target voice to train the sEMG data in both audio and silent modes;

2) Full-A_e: utilizing the electronic sound A_e followed by the subject in silent mode as the target speech for training sEMG data in both audio and silent modes;

3) Silent-A_v: utilizing the recorded voice A_v in audio mode as the target voice to train the sEMG data in silent mode;

4) Silent-A_e: utilizing the electronic sound A_e that the subject reads in silent mode as the target speech to train the sEMG data in silent mode.

We utilized CER and NMOS to assess the comprehensibility and naturalness of speech. Additionally, we employ SMOS and Speaker Verification(SV) accuracy to gauge the similarity between the synthesized speech and the reference speech. The SV accuracy of ground truth speech is to evaluate the speech converted from ground truth speech to VC system.

The evaluation results are shown in Table III. It can be seen that when only using silent sEMG data for training, the synthesized speech quality is worse than that of the model trained with all data, which may be due to the reduction of training data. When using A_e as the target speech, compared to using A_v as the target, the synthesized speech CER of the model trained with all data increased from 4.86% to 6.54%, and the synthesized speech CER of the model trained only with silent sEMG increased from 6.35% to 8.46%. Although the quality of synthesized speech using A_e as the target is inferior to that using A_v as the target, we believe that this degree of decline is acceptable.

This result demonstrates the feasibility of using silent speech mode to collect sEMG signals for training the ETS model. In addition, we also evaluated the speech conversion capability of our WE-ETS system. For the seen cases, the speaker verification(SV) accuracy of all models synthesized speech is higher than 95%. On invisible speakers, the accuracy of SV accuracy drops to about 80% or lower. It can be seen that this result is similar to the result of using ConsistencyVC for speech conversion of ground true audio. This indicates that the vocoder's use of predicted and real Whisper-Embedding features does not disrupt the speaker's embedded features.

The above results indicate that the ETS model can be trained using silent speech mode to collect sEMG signals, and can be restored to the speech of any speaker through a vocoder with speech conversion function. This new training paradigm solves the problem of speech disabled individuals being unable to collect vocal data to train models.

V. CONCLUSION

This study presents WE-ETS, a novel cross-subject sEMG-to-speech conversion system that leverages a pre-trained Whisper model for speaker-independent content feature extraction and adopts diverse calibration strategies to enhance user adaptability. Experimental results demonstrate that WE-ETS consistently outperforms the conventional Direct ETS approach across all metrics in both within- and cross-subject settings, validating the effectiveness of content feature utilization.

Additionally, extensive calibration experiments confirm that the proposed model calibration strategy has strong potential for robust cross-subject adaptation. This study also introduces the first training paradigm utilizing synthesized speech as targets for silent sEMG processing, eliminating reliance on human vocal recordings. These advancements collectively promote the development of more accessible and effective assistive technologies for speech rehabilitation.

VI. FUTURE WORK

While this study demonstrated promising results in sEMG-to-speech conversion using healthy individuals, future work must address the challenges posed by clinical populations, where variations in muscle activation due to atrophy, compensatory strategies, or anatomical changes may affect system performance. Notably, current research efforts have been predominantly based on data from healthy subjects, highlighting the need for broader validation.

Future work will involve recruiting individuals with speech impairments through clinical collaboration to systematically evaluate and optimize the proposed system, ultimately promoting the real-world application of the ETS system as a practical assistive device for voiceless individuals.

DATA AVAILABILITY

Due to the sensitivity of the sEMG and audio data, the dataset cannot be fully anonymized and is therefore not publicly available. Researchers may request access by contacting the corresponding author. Data will be shared upon approval of a signed data-sharing agreement reviewed by the Institutional Ethics Committee, ensuring compliance with privacy regulations and ethical standards.

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